MARYAM REHAN

Cyber Security Internship Report

Week 4 Task 10 What is SIEM? & Install Splunk / Wazuh

Objective

The overall objective of this assignment is to understand the role of Security Information and Event Management (SIEM) in cybersecurity. It focuses on learning how SIEM collects and analyzes security logs, correlates events, generates alerts, and provides centralized security monitoring through dashboards. It also aims to understand the basic functionality, architecture, and benefits of Wazuh as an open-source SIEM platform.

1. Introduction to SIEM

Security Information and Event Management (SIEM) is a cybersecurity solution used to collect, monitor, analyze, and manage security logs from different systems, servers, applications, and network devices.

SIEM helps security professionals identify suspicious activities, detect potential cyber threats, investigate security incidents, and monitor an organization's security environment through a centralized dashboard.

2. What Does SIEM Do?

SIEM performs four main functions:

1. **Log Collection:** Collects security logs from computers, servers, applications, firewalls, and network devices.

2. **Log Correlation:** Connects related security events from different sources to identify suspicious patterns and possible attacks.

3. **Alerting:** Generates alerts when suspicious activities or security rule violations are detected.

4. **Dashboards:** Displays security information, alerts, events, and monitoring statistics in a centralized interface.

3. Three Benefits of SIEM

1. Threat Detection: SIEM identifies suspicious activities, unauthorized access attempts, and potential cyberattacks.

2. Centralized Monitoring: It collects logs from multiple devices and systems in one place, making security monitoring easier.

3. Incident Response: It generates alerts that help security teams investigate and respond to security incidents quickly.

4. How SIEM Works

SIEM works through the following process:

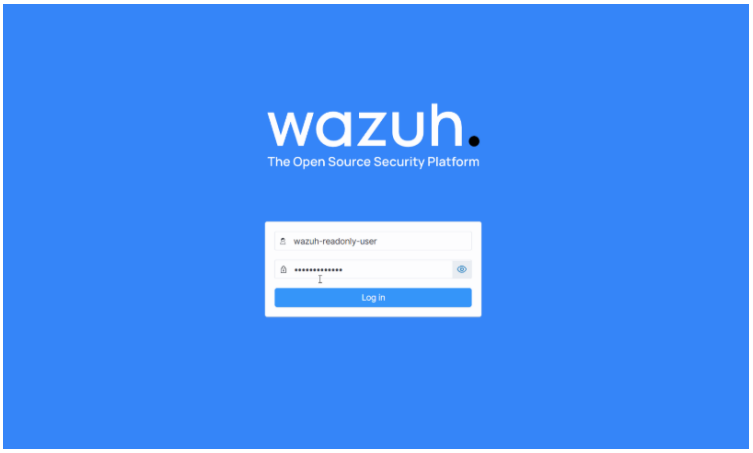
1. Log Collection Collects data from different systems and devices.
2. Log Analysis Examines collected logs to identify unusual activities.
3. Event Correlation Connects related events to detect possible security threats.
4. Alert Generation Generates alerts when suspicious activities are detected.
5. Dashboard Monitoring Displays security events and alerts for security analysts.

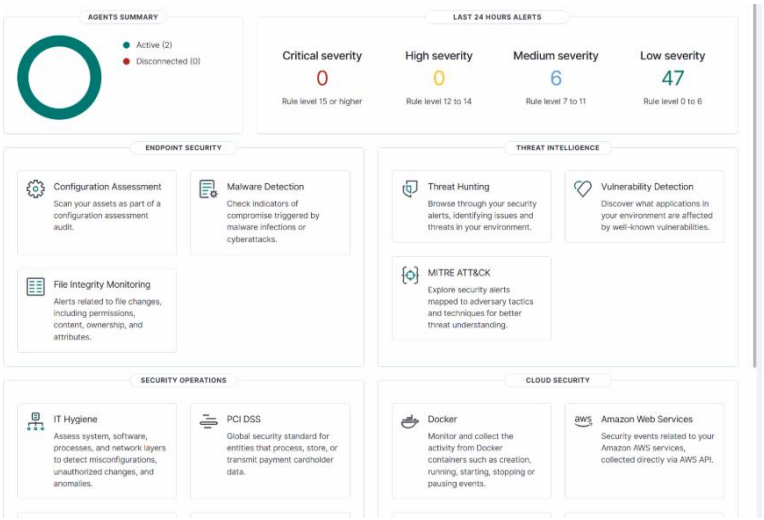
5.Wazuh & Installation Setup Steps

Wazuh is an open-source security monitoring platform used for threat detection, log analysis, vulnerability detection, and security event monitoring. It helps organizations monitor endpoints, detect suspicious activities, and respond to potential security threats.

The basic Wazuh setup process includes the following steps:

1. Open the official Wazuh installation documentation.
2. Review the system requirements.
3. Select the appropriate installation method.
4. Set up the Wazuh server, indexer, and dashboard.
5. Configure Wazuh agents on monitored endpoints.
6. Access the Wazuh dashboard through a web browser.
7. Monitor security events, logs, and alerts.


wazuh.
The Open Source Security Platform
wazuh-readonly-user
Log in


AGENTS SUMMARY
Active (2)
Disconnected (0)
LAST 24 HOURS ALERTS
Critical severity
0
Rule level 15 or higher
High severity
0
Rule level 12 to 14
Medium severity
6
Rule level 7 to 11
Low severity
47
Rule level 0 to 6
ENDPOINT SECURITY
Configuration Assessment
Scan your assets as part of a configuration assessment audit.
Malware Detection
Check indicators of compromise triggered by malware infections or cyberattacks.
File Integrity Monitoring
Alerts related to file changes, including permissions, content, ownership, and attributes.
THREAT INTELLIGENCE
Threat Hunting
Browse through your security alerts, identifying issues and threats in your environment.
Vulnerability Detection
Discover what applications in your environment are affected by well-known vulnerabilities.
MITRE ATT&CK
Explore security alerts mapped to adversary tactics and techniques for better threat understanding.
SECURITY OPERATIONS
IT Hygiene
Assess system, software, processes, and network layers to detect misconfigurations, unauthorized changes, and anomalies.
PCI DSS
Global security standard for entities that process, store, or transmit payment cardholder data.
CLOUD SECURITY
Docker
Monitor and collect the activity from Docker containers such as creation, running, starting, stopping or pausing events.
Amazon Web Services
Security events related to your Amazon AWS services, collected directly via AWS API.

TRY HACK ME



T1

Answer the questions below

What does SIEM stand for?

Security Information and Event Management system ✓ Correct Answer

T2

Answer the questions below

Is Registry-related activity host-centric or network-centric?

host-centric ✓ Correct Answer

Is VPN-related activity host-centric or network-centric?

network-centric ✓ Correct Answer

T3

Answer the questions below

Read the task above.

No answer needed ✓ Correct Answer

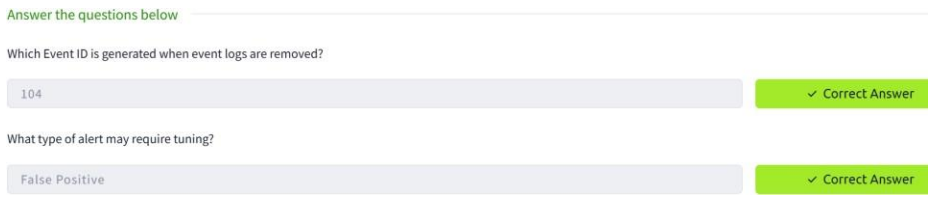
T4

Answer the questions below

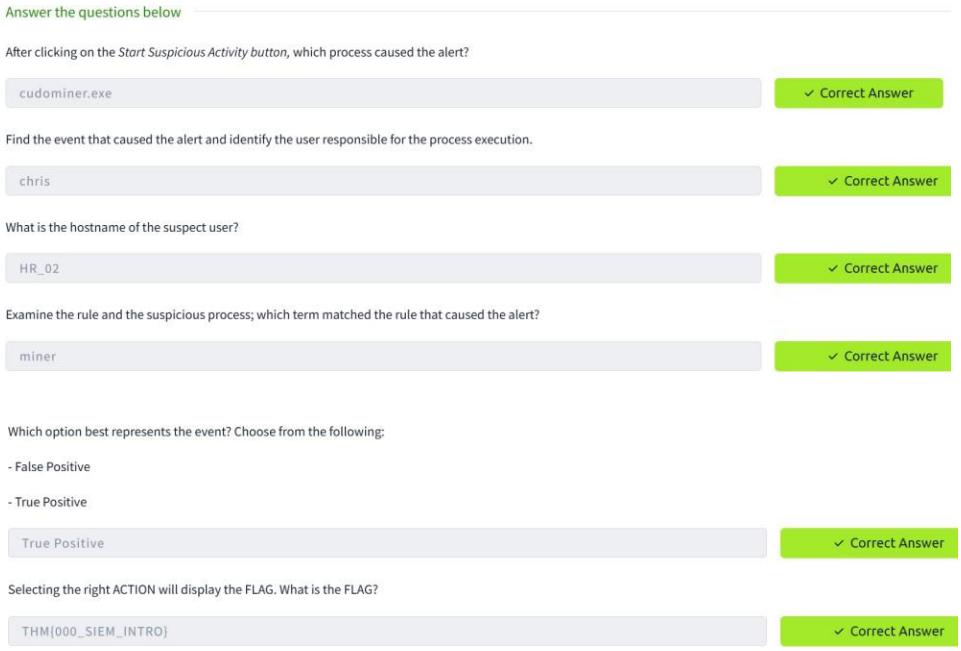
In which location within a Linux environment are HTTP logs stored?

/var/log/httpd ✓ Correct Answer

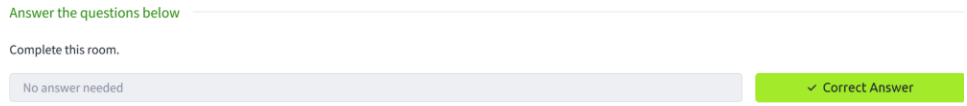
T5



T6



T7



Room completed: 100%

Introduction

This report covers the completion of all seven tasks of the TryHackMe Introduction to SIEM room. It explains the basic concepts of SIEM, security logs, log sources, log ingestion, alert generation, and alert investigation.

Working and Learning

During this room, I learned how Security Information and Event Management (SIEM) collects and monitors security logs from different sources. I understood why centralized log management is important and how SIEM helps security analysts identify suspicious activities and potential threats.

I explored different log sources, including Windows Event Logs, Linux system logs, firewalls, network devices, and applications. I also learned about log ingestion and normalization, which help organize security data for effective monitoring and analysis.

I studied how SIEM detection rules work and how alerts are generated when specific conditions are matched. I learned that Windows Event ID 104 is associated with event log clearing activity, while Event ID 4688 represents process creation activity.

In the practical lab, I investigated a suspicious alert named Potential CryptoMiner Activity using the sample SIEM dashboard. I identified the suspicious process as cudominer.exe, the username as Chris, and the hostname as HR_02. The process matched the detection rule through the term miner.

I classified the activity as a True Positive and selected the action True positive and isolate the host.

TryHackMe FLAG: THM{000_SIEM_INTRO}

Overall Conclusion

I successfully completed all seven tasks of the TryHackMe Introduction to SIEM room with 100% completion. This room improved my understanding of SIEM fundamentals, log collection, log ingestion, centralized monitoring, detection rules, alert generation, and incident investigation.

The practical lab provided experience in identifying suspicious process execution, analyzing event details, recognizing a True Positive alert, and selecting an appropriate security response. Overall, I gained valuable theoretical and practical knowledge of SIEM and its importance in cybersecurity operations.